

Stranding and mass mortality in humboldt penguins (*Spheniscus humboldti*), associated to HPAIV H5N1 outbreak in Chile

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ABSTRACT

The highly pathogenic Avian Influenza virus (HPAIV) H5N1 has caused a global outbreak affecting both wild and domestic animals, predominantly avian species. To date, cases of the HPAIV H5 Clade 2.3.4.4b in penguins have exclusively been reported in African Penguins. In Chile, the virus was confirmed in pelicans in December 2022 and subsequently spread across the country, affecting several species, including Humboldt penguins. This study aims to provide an overview of the incidents involving stranded and deceased Humboldt penguins and establish a connection between these events and HPAIV H5N1. Historical data about strandings between 2009 and 2023 was collected, and samples from suspected cases in 2023 were obtained to confirm the presence of HPAIV H5N1. Between January and August 2023, 2,788 cases of stranded and deceased penguins were recorded. Out of these, a total of 2,712 penguins deceased, evidencing a significant increase in mortality starting in early 2023 coinciding with the introduction and spreading of HPAIV H5N1 in the country. Thirty-seven events were categorized as mass mortality events, with the number of deceased penguins varying from 11 to 98. Most cases (97 %) were observed in the North of Chile. One hundred and eighty-one specimens were subjected to HPAIV diagnosis, four of which tested positive for HPAIV H5N1. Spatial analysis validates the correlation between mass mortality events and outbreaks of HPAIV in Chile. However, the limited rate of HPAIV H5N1 detection, which can be attributed to the type and quality of the samples, requiring further exploration.

1. Introduction

Currently, the Highly Pathogenic Avian Influenza Virus (HPAIV) H5N1 is causing a panzootic that exerts its impact on both, wildlife, and domestic animal populations across the globe. This viral outbreak has generated significant animal mortality, with a particular emphasis on avian species, but also affected a variety of mammal species (Kandeil et al., 2023). In Chile, HPAIV H5N1 was confirmed in December 2022 (Ariyama et al., 2023), serving as a pivotal marker in time. Following this confirmation, the virus rapidly disseminated across the country,

from north to south, impacting a diverse range of animal species. Also, an increase in stranding and mortality events has been observed coinciding with the outbreak of HPAIV H5N1 in Chile, and most species have been confirmed positive for the virus.

The most frequently stranded species reported was the Humboldt Penguin (*Spheniscus humboldti*), whose habitat extends from Isla Foca (5° 12S) in the north of Peru down to Isla Guafo (43° 32S) in the south of Chile. Humboldt Penguins are listed as vulnerable by the IUCN, with a total population of 23,800 mature individuals, 10,200 in Chile and 13,600 individuals in Peru.

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Penguin species are known to be susceptible to Influenza A viruses. Indeed, HPAIV H5 Clade 2.3.4.4b was confirmed in African penguins (Molini et al., 2020). Recognizing the significance of penguins in marine environments and their crucial role in maintaining ecosystem stability (Xavier and Trathan, 2022), they are currently acknowledged as bio-indicators of marine environmental well-being (Barbraud et al., 2020). Understanding the effects of HPAIV H5N1 on penguins is crucial for species preservation, ecological balance, and prevention of zoonotic diseases. Consequently, interventions are necessary to minimize repercussions for penguins, their habitats, other susceptible animal species, and human health.

This study conducts a comprehensive analysis of the stranding and mortality events and their potential link with HPAIV H5N1 on Humboldt penguin populations in Chile during the panzootic outbreak.

2. Materials and methods

2.1. Data collection

The dataset on penguin mortality and stranding was obtained from Chile's Fisheries and Aquaculture Service (SERNAPESCA). It encompasses a time span from January 2009 to August 31, 2023, and essential variables, including date of report, species, location, number of stranded penguins, reproductive status, and whether they were discovered

deceased or alive. Penguins were classified as a stranding (for fewer than 10 birds) or mass stranding (>10 birds) if found in coastal or shallow waters, regardless of their condition (dead, sick, injured, or disoriented) (Fig. 1 and Supplementary information 1). Furthermore, the dataset incorporates supplementary information when available, such as variables on penguin size, body condition, visible lesions, and potential factors that might have contributed to the observed stranding events. In parallel, a dataset of wild birds (except penguins) of HPAIV-confirmed cases, from the same time span, was obtained from Chile's Agricultural and Livestock Service (SAG) for further comparison with the penguin scenario. It is important to note that the number of stranding or mortality events of wild birds was not recorded as we collected only from the penguins.

2.2. Spatial distribution analysis

Spatial distribution analysis of stranding penguin events and the data collected from wild birds was performed to identify spatial high-risk clusters. A high-risk cluster is an area within the study region of a significantly elevated risk for an event such as disease or mortality (Lawson, 2013). Thus, we identified spatial high-risk clusters of stranding of Humboldt penguins and spatial high-risk clusters of positive HPAIV cases in wild birds, with the purpose of visual comparison of overlaps in spatial clusters between both datasets.



Fig. 1. Stranded alive Humboldt Penguin reported in Chile by SERNAPESCA. Penguins were depressed, ataxic, with crusty eyes, and could be easily manipulated by people.

The Kulldorff space exploration statistic was used (Kulldorff, Information Management Services (IMS) (IMS) (IMS), Inc. (2009). Local clusters were assessed using the Bernoulli model of the spatial scan statistic, considering the binary nature of the data (dead = 1, or alive = 0 individuals) for the cluster estimation, all the parameters from the software were used by default and only was considered the search for high-risk clusters.

2.3. Temporal analysis

Time-series analysis was conducted on monthly stranded Humboldt penguin reports, by an additive decomposition model, an autocorrelation test was used to confirm the seasonality of the events, and these results were then analyzed by Ljung-Box test to assess statistical significance. Time series were divided into two, a Holt-Winter forecast that was performed considering data before the 2023 HPAIV outbreak in Chile (2010–2022), to assess for the predicted behavior during 2023 and secondly, a new time series that was generated using the complete data set to visually compare with data observed during 2023. All data management, spatial analysis, temporal analysis, and mapping were performed using Microsoft Excel® 15.12.3, QGIS 3.6 Noosa (QGIS Development Team, 2023), SatScan software version 9.4.2 (Kulldorff, Information Management Services (IMS) (IMS) (IMS), Inc., 2009), and R version 4.0.2 (R Core Team, 2022) and its packages forecast (Hyndman and Khandakar, 2008) and ggplot2 (Wickham, 2011).

2.4. HPAIV molecular detection

Starting on January 2023, after confirmation of HPAIV H5N1 in the avian population in Chile, veterinarians from SERNAPESCA collected samples from stranded penguins to confirm the presence of HPAIV. The standard sampling included tracheal and cloacal swabs. The samples were collected in viral transport media and sent for real-time RT-PCR at the Livestock Virology and Biotechnology Laboratory of SAG. The RNA was obtained using the MagMAX™ Core Viral/Pathogen kit (ThermoFisher, AM1830), and the initial identification was attempted using the VetMAX-Gold AIV Detection Kit (Applied Biosystems™ Catalog Number 4485261) for the generic detection of Influenza A virus (IAV). The subtyping to confirm the presence of H5N1 HPAIV was performed by real-time RT-PCR and Sanger sequencing using the NVSL protocols (1732.02, 1767.01, and 1768.01).

3. Results

3.1. Stranding and mortality of Humboldt penguins in Chile

The overall results of stranding in Humboldt penguins by year are shown in Table 1., where an increase in stranded animals was observed during 2023. Between 2009 and 2022, historical records showed 1035 stranding events involving 1266 Humboldt penguins, with an annual average of 90.4 affected individuals (ranging from 36 to 137). Most of these incidents (979) were single strandings representing 77 % of cases while 6 events involving more than 10 penguins, with the highest count being 61 animals represented 23 % and were classified as mass strandings. Notably, 68 % of the total Humboldt penguins reported stranded previous to year 2023 were found alive.

In contrast, from January to August 2023, there were 907 recorded stranding events of Humboldt penguins, impacting a total of 2788 individuals. Out of these events, 2712 individuals (92 %) were found deceased. When the data observed between the years 2009–2022 is compared to the stranding events observed in 2023, there are statistically significant differences in the dead/alive status evaluated by Chi-square (P value < 0.05).

In 2023, most of live-stranded penguins exhibited a moribund state or distress, characterized by clinical signs such as ataxia, crusty eyes, and unresponsiveness to their immediate environment (Fig. 1). Supplementary Information N°1 shows a video of a Humboldt penguin displaying neurologic clinical signs of the disease.

Out of all cases in 2023, 102 events involved 208 emaciated penguins, while 805 events encompassed 2580 penguins in normal body condition. Additionally, thirty-seven events (4 %) involved more than 11 penguins, with a maximum of 98 individuals in a single instance. These strandings occurred across twelve administrative regions of Chile: Arica y Parinacota, Tarapacá, Antofagasta, Atacama, Coquimbo, Valparaíso, O'Higgins, Maule, Biobío, Ñuble, Araucanía and Los Lagos. The summary of results by month and region can be retrieved from Supplementary Table 1.

3.2. Spatial distribution analysis

Three high-risk clusters for penguin mortality among stranded Humboldt penguins were identified in 2023, all statistically significant (Supplementary Table 2) and located in northern Chile. All penguin high-risk clusters overlapped geographically with HPAIV H5N1 wild bird high-risk clusters reported during 2023 (Fig. 2). Some Humboldt penguin stranding events were also seen between 100 and 150 km from high-risk, statistically significant HPAIV wild bird clusters in the south of Chile.

Table 1
Overall results of Stranded Humboldt Penguin in Chile Between 2009 and August 2023.

Year	Events	N of penguins	Alive	Dead	Undetermined	Regions*
2009	35	50	35 (70.0 %)	15 (30.0 %)	0 (0 %)	II, V, VII, VIII, X, XV
2010	58	80	55 (68.7 %)	11 (13.7 %)	14 (17.5 %)	I, III, IV, V, VII, VIII, X
2011	36	36	35 (97.2 %)	1 (02.7 %)	0 (0 %)	I, IV, V, VII, VIII, X
2012	57	88	52 (59.0 %)	36 (40.9 %)	0 (0 %)	I, IV, V, VII, VIII, X, XI, XV
2013	82	109	67 (61.4 %)	39 (35.7 %)	3 (2.75 %)	I, IV, V, VI, VII, VIII, X, XV
2014	44	46	42 (91.3 %)	04 (8.70 %)	0 (0 %)	II, IV, V, VI, VII, VIII, X, XIV, XV
2015	67	70	58 (82.8 %)	10 (14.2 %)	2 (2.80 %)	II, III, IV, V, VI, VII, VIII, XI, XV
2016	122	128	114 (89.0 %)	14 (10.9 %)	0 (0 %)	I, II, III, IV, V, VI, VII, VIII, X, IX, XIV, XV
2017	106	123	79 (64.23 %)	44 (35.7 %)	0 (0 %)	II, III, IV, V, VI, VII, VIII, X, IX, XIV
2018	57	117	45 (38.4 %)	70 (59.8 %)	2 (1.71 %)	I, II, III, IV, V, VI, VII, VIII, IX, X, XIV, XV
2019	86	96	71 (73.9 %)	25 (26.0 %)	0 (0 %)	I, III, IV, V, VI, VII, VIII, IX, X, XIV, XVI
2020	87	95	68 (71.5 %)	26 (27.3 %)	1 (1.06 %)	I, II, III, IV, V, VI, VII, VIII, X, XIV, XV, XVI
2021	79	91	64 (70.3 %)	27 (29.6 %)	0 (0 %)	II, III, IV, VI, V, VII, VIII, X, IX, XIV, XV
2022	115	137	74 (54.0 %)	37 (27.0 %)	23 (18.98 %)	I, II, III, IV, V, VI, VII, VIII, X, IX, XIV, XV
2023	907	2788	64 (2.30 %)	2712 (97.2 %)	12 (0.43 %)	I, II, III, IV, V, VI, VII, IX, X, XV, XVI

* XV: Arica & Parinacota, I: Tarapacá, II: Antofagasta, III: Atacama, IV: Coquimbo, V: Valparaíso, XIII: Metropolitana, VI: Del Libertador General Bernardo O'Higgins, VII: Maule, XVI: Ñuble, VIII: Bío-Bío, IX: Araucanía, XIV: Los Ríos, X: Los Lagos, XI: Aysén, and XII: Magallanes.

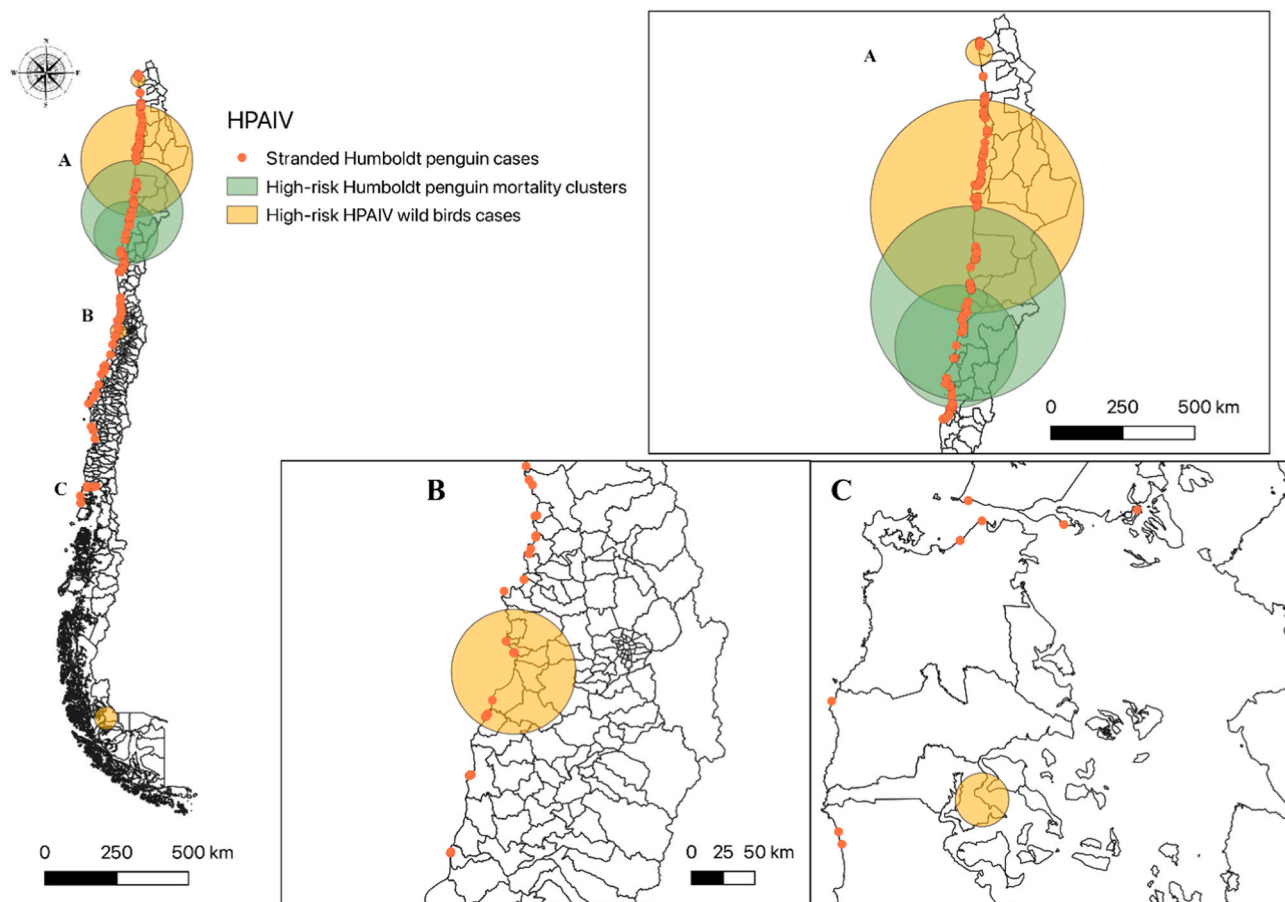


Fig. 2. Spatial distribution of stranded Humboldt penguin (*Spheniscus humboldti*) in the context of the 2023 HPAIV outbreak. A) Spatial overlap between high-risk clusters for Humboldt penguin mortality and wild birds HPAIV confirmed cases in the north of Chile. B) Spatial overlap between stranded Humboldt penguin cases and high-risk HPAIV wild bird clusters in central Chile. C) HPAIV wild bird clusters and middle distance (between 100 and 150 km) between stranded Humboldt penguin cases in southern Chile.

The temporal analysis of all stranding events prior to 2023 showed a regular pattern, the Holt-Winters forecast was performed for the data until December of 2022 showing the same behavior, which contrasts with the result from 2023 (Fig. 3). Seasonality was observed in the additive decomposition (Fig. 4), during the Chilean summer season (21st December – 20th March) and confirmed by the Ljung-Box test, showing statistically significant non-zero autocorrelation values ($p = 6.217 \times 10^{-15}$). This suggests that 2023 stranding and mortality events don't align with the regular pattern observed for prior events, in both seasonality and frequency, and are a result of different causes occurring specifically in 2023 in Chile.

3.3. Detection of HPAIV H5N1

Out of the 181 cases corresponding to Humboldt penguin submitted for HPAIV diagnostic, four cases (2 %) tested positive for generic IAV and confirmed positive for HPAIV H5N1. The positive samples were collected from the Northern regions of Tarapacá, Antofagasta, and Coquimbo and exhibited RT-PCR cycle threshold (Ct) values ranging from 24 to 28. Three out of the four positive samples were tracheal swabs, and one was a pool of both cloacal and tracheal swabs collected from the same animal. The analysis confirmed that all positive samples belong to the H5 subtype within clade 2.3.4.4b.

4. Discussion

The global impact of HPAIV H5N1 has highlighted its threat to

biodiversity and animal health, particularly in avian species. The recent confirmation of the HPAIV H5 Clade 2.3.4.4b virus in Chile (Ariyama et al., 2023), has extended its impact to various animal species. In this report, we confirmed the HPAIV H5 Clade 2.3.4.4b in Humboldt penguins and aimed to establish a link between the increasing incidents of stranded and dead penguins and the virus.

In 2023, in only 8 months, there was a significant increase of stranding in Humboldt penguins > 30 times compared to a regular year. Of the total stranded penguins, 97 % were found dead, which is notably uncommon. From the scarce number of live stranded penguins, it was possible to observe clinical signs compatible with the influenza virus infection including neurological disease (Fig. 1, Supplementary information 1). Additionally, it was confirmed that swab samples collected from the cloacal and tracheal regions of the Humboldt penguins revealed the presence of the HPAIV H5 Clade 2.3.4.4b in 4 out of 181 individuals sampled. This is the first report of HPAIV H5N1 confirmed in penguins in South America. It is well known that penguins are susceptible to the Influenza A virus, and there is evidence of that, especially in those inhabiting Antarctica which has been most studied. Since 1981, free-range penguins from Antarctica, *Pygoscelis adeliae*, *Pygoscelis antarctica*, and *Pygoscelis papua*, have tested positive for H7Nx, H1N1, H3Nx, H9Nx by antibody detection methods. More recently, H11N2 and LPAIV H5N5 have been isolated and characterized in the same species (Hurt et al., 2014, 2016; Barriga et al., 2016). However, the information about influenza A virus from South American penguins such as Humboldt penguins is scarce. Smith, et al., 2008 studied 98 Humboldt penguins in Peru without detecting IAV antibodies, similar results were

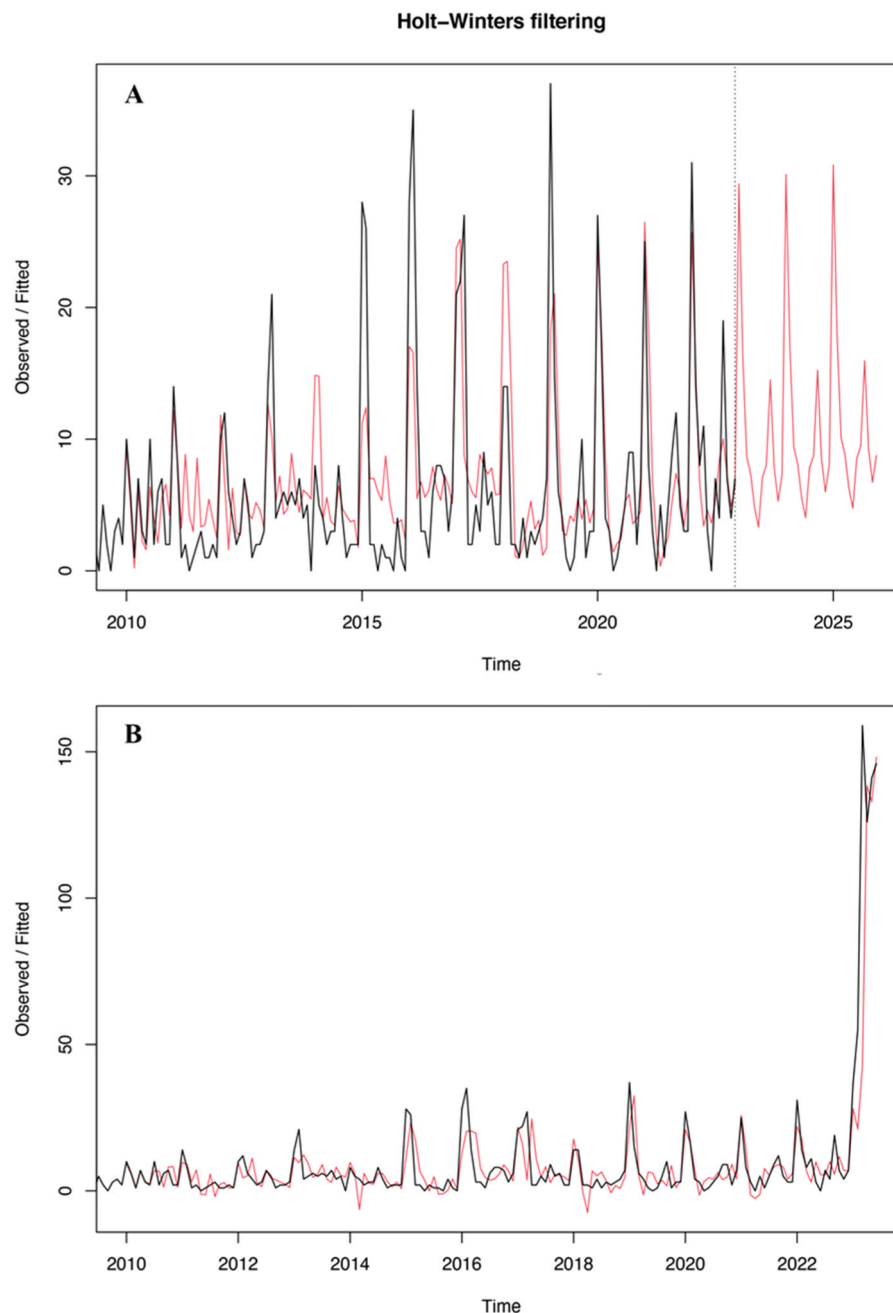


Fig. 3. Time series and Holt-Winters forecast of stranded events for Humboldt penguin (*Spheniscus humboldti*). The black line represents the original data, red line represents the forecasted time series. A) Holt-Winters forecast for stranded events with data until December 2022, including predicted values until 2025. B) Holt-Winters forecast for stranded events until July 2023.

obtained by [Uhart et al. \(2020\)](#), those who did not detect IAV antibodies in more than 1000 Magellanic penguins collected in Argentina between 1994 and 2008. In contrast, HPAIV H5 clade 2.3.4.4b has been extensively reported in African penguins since 2017 [Peyrot et al. \(2022\)](#) and [Abolnik et al. \(2023\)](#) reported both subtypes H5N8 and H5N1 (H5) clade 2.3.4.4b HPAIV in African penguins which were related to mass die-offs in this species as well as other avian species. More recently, [Leguia et al. \(2023\)](#) also observed mass die-offs of Humboldt penguins on the Peruvian coast confirming the virus in several other species, although the samples from penguins tested negative. As evidenced in this article all recent reports have linked HPAIV to mass die-offs in penguins as evidenced in this paper.

[Pardo-Roa et al., 2023](#) ([Pardo-Roa et al., 2023](#)) sequence more than 100 viruses from the Chilean outbreak including two out of the four

positive samples, the (A/Humboldt penguin/Tarapaca/238744-2/2023 (H5N1)) and (A/Humboldt penguin/Coquimbo/239590/2023 (H5N1)) (GenBank accession numbers OR125167- OR269265, respectively). These viruses were independently related to viruses sequenced from different avian species, suggesting that at least, there was initially direct transmission from other avian species instead of penguin-to-penguin transmission. A possible source of infection to penguins may be explained through the contamination of the marine environment with the HPAIV coming from secretions of other wild sea bird carriers when sharing a common feeding habitat. The latter has been explained and validated as the viral route of transmission from wild sea birds to mammalian species ([Ulloa et al., 2023](#)).

Historically, fluctuations in Humboldt penguin populations have been primarily attributed to the El Niño southern oscillation (ENSO)

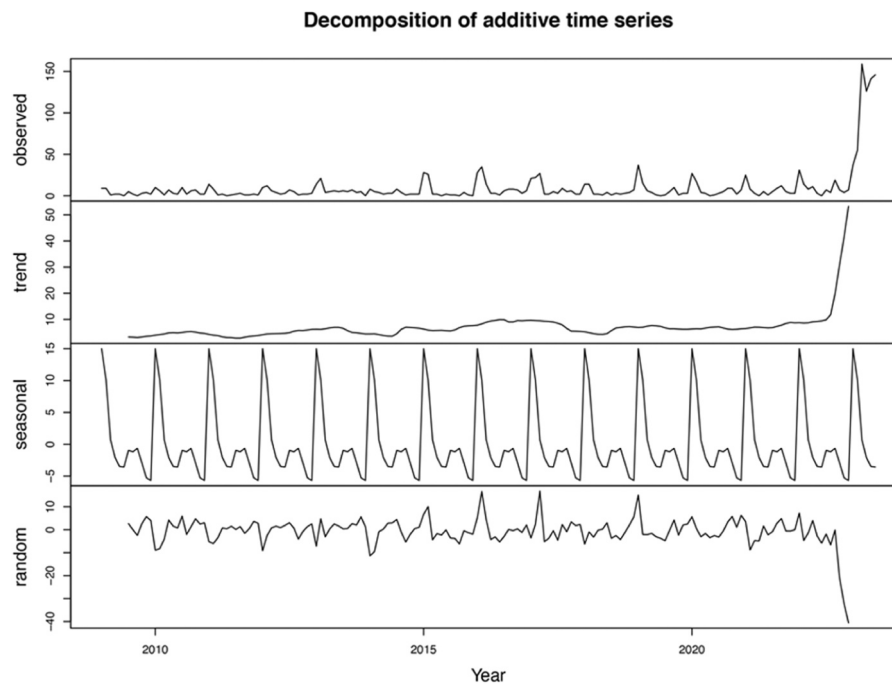


Fig. 4. Seasonal additive decomposition for stranded events of Humboldt penguin (*Spheniscus humboldti*) in Chile, during 2009–2023. The seasonal component shows a regular pattern, random and trend component shows a regular pattern until 2023 in accordance with the HPAIV outbreak in Chile.

(Culik et al., 2000), which disrupts normal sea conditions. ENSO has a significant impact on penguin populations, resulting in substantial declines, especially among young penguins, who represented 71 % of the cases during events of mass mortality in 1983 (Hays, 1984). In contrast, in this study, 76 % of stranded individuals were adults aligning with Molini et al., (2020), during the H5N8 outbreak in African penguins, where almost all of the observed dead penguins were in adult plumage. Additionally, during ENSO, adult penguins underwent noticeable physical changes, with their weight dropping significantly to approximately 1.9–3 kg, compared to their regular weight of 5 kg (Hays, 1984). In the current study, nearly 92 % of penguins were found in a normal body condition, indicating not only the presence of disease with an acute clinical course in this species but also a substantial deviation from the conditions observed during ENSO. Moreover, another noteworthy contributor to the mass mortality of penguins in Chile is bycatch in artisanal fishing, although these events tend to be more localized and acute in their impact (Schalter et al., 2009). Another worthwhile point to be considered on this respect is that diagnoses consistent with net capture include wing fractures, skin contusions, and net marks besides hemorrhages of subcutaneous tissue at the neck and shoulders all external signs easy to be recognized by trained personnel without even performing a necropsy for differential diagnosis (Hamel et al., 2009).

In addition, our study demonstrates a spatial correlation between the detection of stranded and dead penguins and other positively confirmed avian influenza cases, supporting the connection between these mass mortality events and HPAIV outbreaks. While our study provides valuable insights into the impact of HPAIV H5 Clade 2.3.4.4b on Humboldt penguins, it is essential to acknowledge its limitations. Firstly, the low rate of positive virus detection raises questions about sample quality and the timing of sample collection. Many samples from dead animals were collected without precise documentation of their date of death, often hours after the penguins had deceased. Additionally, since these animals were found on the coast, there is a potential for temperature fluctuations that could have impacted result accuracy, considering the known effect of temperature on the persistence and transmission of HPAIV (Lowen et al., 2007, 2008). In Northern coastal Chile, temperatures range from 18°C to 25°C throughout most of the year, whereas in the southernmost

coastal regions of the species distribution, temperatures range from 7°C to 17°C. Fluctuations in environmental temperature can affect the persistence of the virus and thus the integrity and sensitivity of the tests (Ma et al., 2009; Martin et al., 2018). Furthermore, the limited number of positive samples may hinder a comprehensive understanding of the pathogenesis in penguin populations. Improving sampling methods and conducting further research into the complex dynamics of HPAIV in penguins are essential steps for a more thorough understanding of this intricate disease. This is particularly crucial given the estimated population of approximately 10,200 Humboldt penguins in Chile, classified as vulnerable by the International Union for the Conservation of Nature (IUCN) in 2020 (BirdLife International., 2020) raising concerns about their risk of extinction. Their potential disappearance could have far-reaching consequences, including disruptions in marine ecosystems, predator-prey dynamics, and broader impacts on biodiversity (Xavier and Trathan, 2022). It could also signal environmental degradation with adverse effects on coastal and oceanic ecosystems (Campbell et al., 2019). The high mortality rate of 20.6 % due to the HPAIV outbreak emphasizes the urgent need for targeting conservation efforts to protect the species and its ecological role. Furthermore, these challenges may extend to other penguin species facing similar environmental pressures, highlighting global implications for penguin conservation. Additionally, Chile is home to a diverse penguin population, with 18 distinct species most of them (89 %) in the Antarctic territory (Vianna et al., 2020). Notably, there have also been mass strandings involving Magellanic penguins, with 546 individuals affected (16 % of the total stranded penguins). However, the impact on Magellanic penguins is less severe than what has been observed among Humboldt penguins. One possible explanation for this difference lies in the geographic distribution of these two species. Magellanic penguins primarily inhabit southern regions, from Coquimbo (29° latitude S) to Cape Horn (56° latitude S), while Humboldt penguins are found from Foca Island (5° latitude S) off the coast of Peru to Puñihuil Islands (42° latitude S) in Chile. This geographic distinction aligns with the virus's origin in northern Chile in December and its subsequent southward spread, which coincides with the timing of the study. Furthermore, it is important to bear in mind the potential epidemiological risk of this scenario considering that it has

been described that the Humboldt penguin has the ability to migrate up to 895 km (Culik et al., 2000), potentially giving rise to new variants and increasing the probability of spreading the virus.

To date, the HPAIV H5N1 has been detected in fulmars on Falkland Island (Bennison et al., 2023), thereby elevating the risk of further dissemination through migratory birds to countries where the virus has not been detected to date, such as Australia and New Zealand.

Furthermore, it would be beneficial to utilize tissue samples such as the lungs and brain in these studies. This approach can provide insights into the specific physiological mechanisms underlying the increased susceptibility of penguins compared to other species, allowing for a more comprehensive understanding of the disease's impact on both wildlife and public and animal health. The potential presence of HPAIV H5N1 among Humboldt penguins' population is a concerning issue with multiple dimensions. It is not only associated with penguin mortality but also poses risks of spreading to Antarctica and neighboring countries through migratory bird routes. Comprehensive research, conservation efforts, and international cooperation are essential to address these interconnected challenges effectively and safeguard both avian populations and global health.

CRediT authorship contribution statement

Victor Neira-Ramirez: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Hugo Araya:** Resources, Funding acquisition, Conceptualization. **Mauricio Ulloa:** Writing – original draft, Validation, Supervision, Conceptualization. **Gabriela Muñoz:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carolina Aguayo:** Methodology, Investigation, Formal analysis. **Magdalena Johow:** Investigation, Funding acquisition, Formal analysis, Data curation. **Joaquín Atavales:** Methodology, Investigation, Formal analysis. **Barbara Quezada:** Validation, Software, Methodology. **Raúl Alegría:** Visualization, Software, Methodology, Investigation. **Nikita Enciso:** Methodology, Investigation, Formal analysis. **Benjamín Bennet:** Writing – review & editing, Resources, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Data Avail ability statement

All the data that support this study are available upon request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.prevetmed.2024.106206](https://doi.org/10.1016/j.prevetmed.2024.106206).

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